

A Safety Boundary Governor for Teleoperated Robotic Haircutting

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Abstract—Robotic haircutting requires a cutting tool to operate close to a person’s head, where small command errors may violate strict spatial safety constraints. Existing haircutting demonstrations have shown functional feasibility, but reported safety mechanisms and quantitative validation remain limited. This paper proposes a head-scan-based safety boundary governor for teleoperated robotic haircutting. The method combines a 3D scan of the head with a spherical tool-envelope model and regulates boundary-crossing reference commands before they are sent to the robot. When a commanded reference enters the configured boundary, the governor projects the executable command to the safety boundary and removes the inward normal component while preserving tangential motion and retreat from the head. To validate the approach, experiments were conducted in safe-input, no-governor baseline, and boundary-interaction conditions using a mannequin-head setup. Across five boundary-interaction trials, the prescribed reference input intentionally entered the configured boundary, with a maximum boundary-entry depth of 153.5 mm and 50.3% of samples inside the boundary, whereas the governed command and measured robot output, obtained from robot controller feedback, had no inside-boundary samples. These results suggest that the proposed governor provides a practical robot-side command-regulation layer for robotic haircutting teleoperation in the tested setup.

Index Terms—Robotic haircutting, teleoperation, human-robot interaction, safety governor, near-head manipulation

I. INTRODUCTION

As service robots increasingly enter personal-care scenarios, robotic haircutting has become a representative task involving close-range human-robot interaction. Unlike general service manipulation, haircutting requires a robot to deliberately move a cutting tool close to the human head and perform local trimming along a person-specific head contour [1]. Therefore, the robot cannot simply keep away from the human body, nor can it rely only on a static safety distance to avoid risk. The central difficulty is that the tool must remain controllable near the head boundary, while any command error directed toward the inside of the head may turn from a normal trajectory-tracking error into a safety violation [2], [3]. Existing robotic haircutting demonstrations and surveys have discussed the feasibility

and challenges of this task from the perspectives of service robotics, automation, mobile robotics, perception, planning, hair deformation, human motion, and physical human-robot interaction [4]–[6]. Among these challenges, runtime safety control is a fundamental requirement that must be addressed before robotic haircutting systems can move toward practical use [7].

Existing robotic haircutting systems have mainly demonstrated functional feasibility. As shown in Table I, reported systems include automatic trimming demonstrations on real human subjects and a proof-of-concept safety framework based on a three-dimensional scan of a mannequin head. These studies demonstrate the potential of robotic haircutting, but runtime algorithm-level safety constraints and trajectory-level quantitative validation remain limited. For a practical haircutting scenario, setting a static safety margin or triggering an emergency stop only after a boundary violation is not sufficient. The system must determine before execution whether a commanded reference would cause the tool envelope to cross the head safety boundary and, when necessary, transform the boundary-crossing reference into a feasible executable command. At the same time, this transformation must not simply block all motion near the boundary, because tangential motion along the head boundary and retreating motion away from the head are themselves necessary for haircutting.

A related line of work in surgical robotics addresses tool safety through virtual fixtures and active constraints. These methods typically restrict surgical tool motion with respect to anatomy-derived constraints, forbidden regions, or no-fly zones to improve operational safety [10]–[12]. This principle is relevant to robotic haircutting, but the task conditions are different. First, the safety boundary in robotic haircutting is usually obtained from externally visible, person-specific head geometry rather than internal anatomical structures in preoperative medical images. Second, a haircutting tool is not an ideal point but has a finite distal envelope; therefore, the safety check should be applied to the distal tool envelope rather than to a single tool center point (TCP).

TABLE I
COMPARISON OF REPRESENTATIVE ROBOTIC HAIRCUTTING SYSTEMS AND THE PROPOSED SAFETY-BOUNDARY GOVERNOR.

Work	Category	Scenario	Head modeling method	Safety mechanism	Quant. validation	Ref.
Buzz Robotics	Automatic system	Real human	Vision-based modeling	Not reported	✗	[8]
Stuff Made Here	Automatic system	Real human	Not reported	Mechanical distance measurement	✗	[9]
Safety in Robotic Haircutting	Safety PoC	Mannequin head	3D scan	Three-layer safety framework	△	[7]
This paper	Teleoperation system	Mannequin head	3D scan	Boundary governor with command projection and normal-component filtering	✓	–

Remark: ✓ indicates quantitative safety validation with reported trajectory-level statistics; △ indicates proof-of-concept safety testing without comparable trajectory-level statistics; ✗ indicates that quantitative safety validation was not reported.

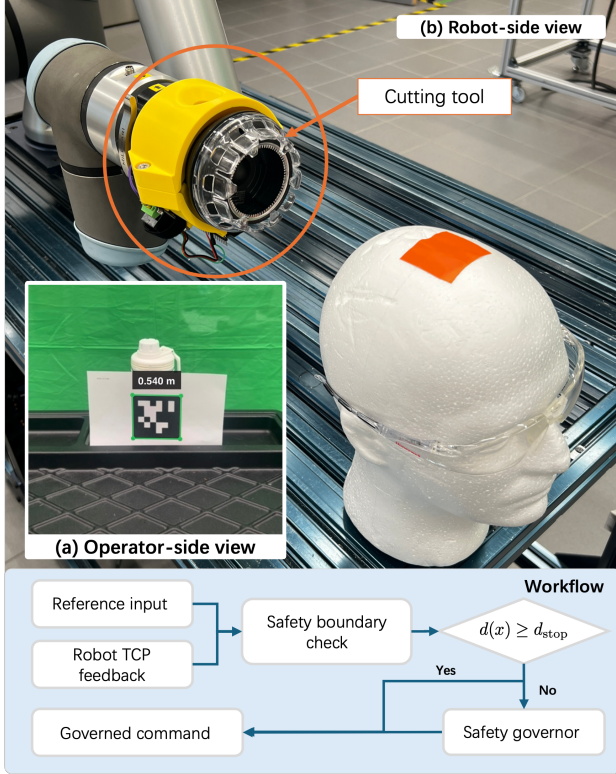


Fig. 1. Experimental setup and command-regulation workflow of the teleoperated robotic haircutting system.

More importantly, in haircutting, approaching and moving along the head boundary is not an abnormal condition but part of the task. Robotic haircutting therefore requires not a simple collision-avoidance or command-rejection mechanism, but an algorithm-level safety regulation method that preserves useful boundary-following motion while removing inward-penetrating motion components.

This paper studies this problem in a teleoperated robotic haircutting scenario. Teleoperation is adopted because, at the current stage, local tool placement, hairstyle intention, and task adaptation still benefit from human judgment, whereas the robot can provide precise motion execution and runtime safety enforcement. The experimental setup is shown in Fig. 1. The operator-side client estimates the handheld tool pose and sends

tool commands to the robot-side control server. The server converts each received teleoperation command into a robot TCP target and applies a head-scan-based safety boundary constraint before execution. In boundary-probing motions, the reference can intentionally cross the boundary threshold to test whether the robot-side governor prevents the executable command from doing so. Therefore, this paper focuses on the following problem: how to perform real-time geometric safety regulation for each near-head motion command on the robot side without changing the operator input reference or the teleoperation interface.

To address this problem, this paper proposes a head-scan-based safety-boundary governor for near-head command regulation in teleoperated robotic haircutting. The method uses the three-dimensional scan of a mannequin head as a scan-specific safety boundary and approximates the occupied space of the distal cutting tool using a spherical envelope. Before each teleoperation command is executed, the governor checks whether the reference tool envelope would cross the boundary threshold. If the reference is already feasible, it is passed through unchanged. If the reference crosses the threshold, it is projected onto the safety boundary, and only the inward normal component of the motion is removed. As a result, the system prevents the executable tool command from advancing into the head while preserving tangential motion along the boundary and retreating motion away from the head. The proposed mechanism is implemented as a robot-side geometric safety layer in the teleoperation pipeline, without changing the operator interface or the absolute command reference. The system is evaluated on a mannequin-head platform under safe-input, no-governor baseline, and boundary-interaction conditions by comparing the prescribed reference input, the governed command, and the measured robot trajectory.

II. METHODOLOGY

This section describes the robot-side safety governor used to convert the operator's teleoperation input into a command that satisfies the head-centered safety boundary. The method is built around two geometric models: a convex boundary derived from the 3D head scan and a spherical envelope representing the distal cutting tool. Based on these models, the governor first checks whether a reference command is feasible, then

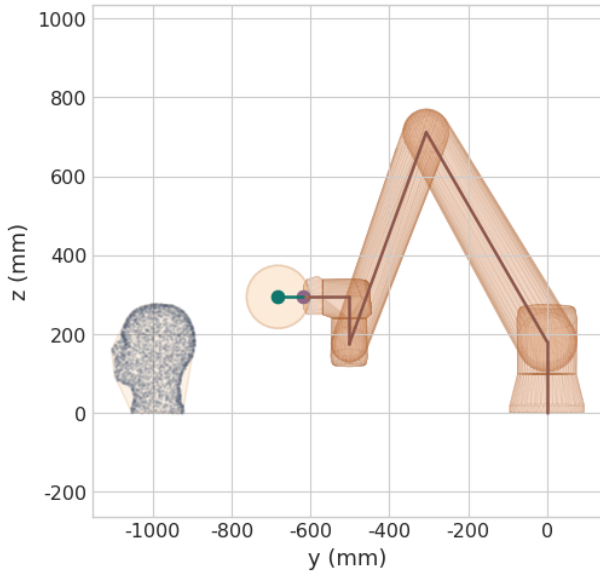


Fig. 2. Robot-side geometric model used by the safety-boundary governor.

applies position projection and normal-component filtering when the reference would cross the boundary threshold.

At each control cycle k , the teleoperation pipeline provides a desired tool center point (TCP) pose, denoted by $(p_k^{\text{des}}, R_k^{\text{des}})$, where $p_k^{\text{des}} \in \mathbb{R}^3$ is the desired TCP position in the robot base frame and $R_k^{\text{des}} \in SO(3)$ is the desired TCP orientation. This reference pose represents the operator's intent and is kept in the absolute hand-to-TCP reference frame. The safety governor computes a corrected TCP position command $p_k^{\text{cmd}} \in \mathbb{R}^3$ that can be sent to the robot without allowing the tool envelope to cross the head boundary threshold. The TCP orientation is not modified by the governor and is kept as R_k^{des} . The correction is decomposed into two operations: first, projection of a boundary-crossing tool-envelope center back to the safety boundary; second, filtering of the commanded displacement so that only the inward normal component is removed.

Projection to the safe set. The head scan is first represented by a convex safety boundary in the head frame. Each boundary face is written as a normalized half-space

$$n_i^\top x + b_i \leq 0, \quad i = 1, \dots, N, \quad (1)$$

where x is a point in the head frame and n_i is the outward unit normal of face i , b_i is the scalar offset of that face, and N is the number of boundary faces. The cutting tool is not treated as a point. Instead, the distal tool geometry is approximated by a spherical envelope of radius r_t , whose center has a fixed offset q_t from the TCP. For the desired TCP command $(p_k^{\text{des}}, R_k^{\text{des}})$, the desired envelope center expressed in the head frame is

$$x_k^{\text{des}} = R_h^\top (p_k^{\text{des}} + R_k^{\text{des}} q_t - t_h), \quad (2)$$

where (R_h, t_h) is the head pose in the robot base frame. For any envelope center x , the active face is the one with the largest boundary value:

$$a = \arg \max_i (n_i^\top x + b_i). \quad (3)$$

For this active face, define $\sigma(x) = n_a^\top x + b_a$. The active-plane boundary distance of the tool envelope is then $d(x) = \sigma(x) - r_t$. This value is used by the governor as the local boundary distance. Thus the feasible set for the envelope center is

$$\mathcal{S} = \{x \mid d(x) \geq d_{\text{stop}}\}, \quad (4)$$

where d_{stop} is the prescribed minimum allowed boundary distance for the tool envelope. Equivalently, $\sigma(x) \geq m$, with $m = r_t + d_{\text{stop}}$. For an infeasible candidate with $\sigma(x) < m$, the governor moves it along the active outward normal. Let $x' = x + \alpha n_a$. Because the normal is unit length,

$$\sigma(x') = \sigma(x) + \alpha. \quad (5)$$

The projected point should lie exactly on the safety threshold, i.e., $\sigma(x') = m$. Therefore $\alpha = m - \sigma(x)$, which gives the projection for the infeasible candidate

$$\Pi(x) = x + (m - \sigma(x)) n_a, \quad (6)$$

and, consequently, $d(\Pi(x)) = d_{\text{stop}}$. This step is a position-level projection: it maps a boundary-crossing envelope center to the nearest active-plane threshold along the local outward normal. The robot-side geometric model used by the governor is illustrated in Fig. 2.

Normal-component filtering. The governor preserves the absolute teleoperation reference by correcting only the commanded target. The desired position x_k^{des} is still computed from the absolute hand-to-TCP mapping, even when it lies inside the boundary threshold. Let $\bar{x}_{k-1}$ be the previous governed envelope center, after projection by $\Pi(\cdot)$ if needed. The displacement requested by the operator from the last governed command to the current desired point is

$$\Delta_k = x_k^{\text{des}} - \bar{x}_{k-1}. \quad (7)$$

Let $a_k = \arg \max_i (n_i^\top \bar{x}_{k-1} + b_i)$ and use the corresponding local outward normal $n_k = n_{a_k}$ for filtering. The scalar $\Delta_k^\top n_k$ is the component of the requested displacement along this normal. A positive value moves the tool away from the head, a zero value moves tangentially, and a negative value moves the tool inward. The safe displacement is defined as

$$\Delta_k^{\text{safe}} = \begin{cases} \Delta_k, & x_k^{\text{des}} \in \mathcal{S}, \\ \Delta_k - \min(\Delta_k^\top n_k, 0) n_k, & x_k^{\text{des}} \notin \mathcal{S}. \end{cases} \quad (8)$$

For an infeasible reference, if $\Delta_k^\top n_k < 0$, the term $(\Delta_k^\top n_k) n_k$ is precisely the inward normal component and is subtracted. If $\Delta_k^\top n_k \geq 0$, the minimum is zero and the requested motion is kept unchanged. Thus the governor permits sliding along the boundary and immediate retreat from the head, but suppresses further inward motion.

For $\|\Delta_k^{\text{safe}}\| > 0$, the filtered step is bounded by the release limit $\rho > 0$, which specifies the maximum allowed envelope-center displacement per control cycle:

$$\tilde{\Delta}_k = \min(1, \rho / \|\Delta_k^{\text{safe}}\|) \Delta_k^{\text{safe}}. \quad (9)$$

If $\|\Delta_k^{\text{safe}}\| = 0$, then $\tilde{\Delta}_k = 0$. The resulting candidate envelope center is

$$\tilde{x}_k = \bar{x}_{k-1} + \tilde{\Delta}_k. \quad (10)$$

TABLE II
BOUNDARY-GOVERNOR PERFORMANCE AND NO-GOVERNOR BASELINE.

Trial	Duration (s)	Reference worst/inside (mm/%)	Command worst/inside (mm/%)	Measured worst/inside (mm/%)	Intervention (%)	Mean tracking error (mm)
1	50.8	-54.3 / 27.7	0.6 / 0.0	1.9 / 0.0	27.7	7.3
2	38.9	-42.5 / 14.8	0.1 / 0.0	1.3 / 0.0	14.8	7.6
3	72.4	-135.5 / 53.2	0.2 / 0.0	1.1 / 0.0	53.2	10.3
4	60.4	-99.6 / 54.2	0.1 / 0.0	0.7 / 0.0	54.2	6.5
5	63.0	-153.5 / 83.4	0.2 / 0.0	3.2 / 0.0	83.5	10.4
All boundary	285.5	-153.5 / 50.3	0.1 / 0.0	0.7 / 0.0	50.3	8.6
No governor	25.6	-140.3 / 39.7	-140.3 / 39.7	-140.1 / 39.9	0.0	13.0

Remark: Reference/Command/Measured report worst distance and inside-boundary percentage. Negative values indicate boundary entry. Intervention is the percentage of modified samples; tracking error is the measured-to-command TCP error.

If $\tilde{x}_k \notin \mathcal{S}$, scalar bisection on $[\tilde{x}_{k-1}, \tilde{x}_k]$ selects the farthest feasible point. Let x_k^{cmd} denote the final feasible envelope center in the head frame, i.e., $x_k^{\text{cmd}} = \tilde{x}_k$ if $\tilde{x}_k$ is feasible and the bisection result otherwise, and let $g_k^{\text{cmd}} = R_h x_k^{\text{cmd}} + t_h$ be the corresponding envelope center in the robot base frame. The final TCP command is $p_k^{\text{cmd}} = g_k^{\text{cmd}} - R_k^{\text{des}} q_t$, and the logged correction is $e_k^{\text{sat}} = \|p_k^{\text{des}} - p_k^{\text{cmd}}\|$, which quantifies the safety intervention without re-anchoring the teleoperation reference. After each cycle, the governed envelope center is updated as $\tilde{x}_k = x_k^{\text{cmd}}$.

III. EXPERIMENTAL VALIDATION AND RESULTS

This section evaluates the proposed governor in logged teleoperation trials and reports both representative trajectories and aggregate safety statistics. The validation is organized into the experimental setup, which defines the boundary model, tool envelope, test conditions, and metrics, followed by the results, which compare the prescribed reference input, governed command, and measured robot motion.

A. Experimental Setup

The validation tests whether the proposed safety governor can prevent the robot-side tool command from entering the head safety boundary while preserving the operator's absolute teleoperation input as the reference. It uses logged teleoperation trials from the robotic haircutting setup in Fig. 1. The head scan is represented by the convex safety boundary used in Sec. II, and the distal cutting tool is approximated by a spherical envelope with radius $r_t = 80$ mm and TCP-frame offset $q_t = [0, 0, 64]^T$ mm. The safety threshold is set to $d_{\text{stop}} = 0$ mm, and the release limit is set to $\rho = 100$ mm for all trials. In this validation, safety is evaluated with respect to the reconstructed head-boundary model. A larger positive threshold can be selected in practical deployment to account for scan, calibration, and tracking uncertainties.

For each logged sample, three trajectories are evaluated against the same boundary model: the prescribed reference input, the governed command, and the measured robot output. The measured robot output is obtained from the robot controller feedback and evaluated using the same boundary model. The recorded trials include one safe-input condition,

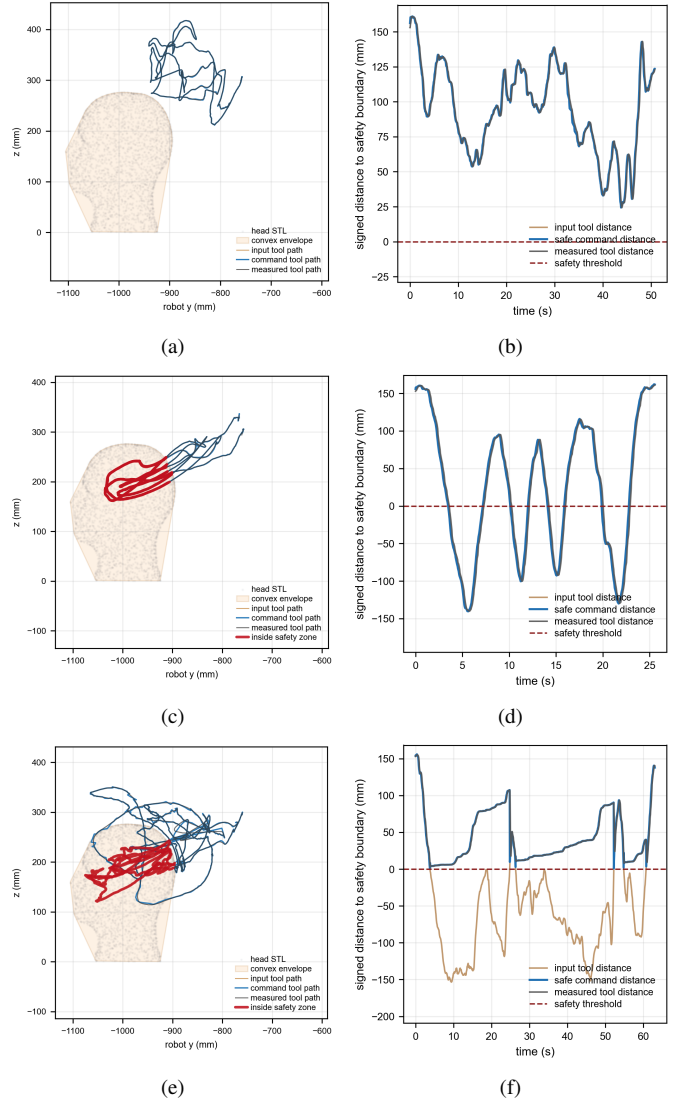


Fig. 3. Representative safety-boundary validation trials. Left: spatial trajectory overlays. Right: reference, command, and measured boundary distances over time.

in which the operator reference remains outside the boundary; one no-governor baseline condition, in which the same type

of boundary-challenge reference is executed without command regulation; and five boundary-interaction trials with the governor enabled. The logged trials were sampled at approximately 26.6 Hz, and the reported metrics are the worst boundary distance relative to the configured boundary, the percentage of samples inside that boundary, the intervention ratio, and the Euclidean tracking error between the measured and logged command TCP positions. The intervention ratio is defined as the percentage of samples for which the governor modifies the reference command before execution. In Table II, each trajectory column reports the worst boundary distance followed by the percentage of inside-boundary samples. The command column denotes the logged governor command in the boundary-interaction trials and the logged ungoverned command in the no-governor row. Across the five boundary-interaction trials, the summary values reported in the text use the worst boundary distance, total duration, sample-weighted percentages, and mean tracking error.

B. Results and Discussion

Fig. 3 shows the representative behavior of the safety governor under the three test conditions. In the safe-input condition, the reference, governed, and measured trajectories remain close to each other and stay outside the configured boundary, indicating that the governor does not introduce correction when the operator command already satisfies the boundary constraint. In the no-governor baseline condition, the executed trajectory enters the configured boundary, confirming that the boundary-challenge reference is an effective stress test for the safety layer. With the boundary governor enabled, whenever the reference input enters the configured boundary, the command is regulated to remain outside the boundary, and the measured output also stays outside the boundary.

Across the five boundary-interaction trials, the same behavior was observed over 285.5 s of logged operation. The prescribed reference input intentionally entered the configured boundary, with a maximum boundary-entry depth of 153.5 mm, corresponding to a worst boundary distance of -153.5 mm, and 50.3% of samples inside the boundary. In contrast, the governed command and measured output had no inside-boundary samples; their worst boundary distances were 0.1 mm and 0.7 mm, respectively. The measured-command tracking error was 8.6 mm on average. The no-governor baseline produced a maximum measured boundary-entry depth of 140.1 mm, corresponding to a worst boundary distance of -140.1 mm, and had 39.9% inside-boundary samples. These results show that the governor regulated boundary-crossing reference commands at the algorithm level and that the robot-side measured motion respected the head-scan safety boundary within the tested setup.

IV. CONCLUSION

This paper presented a head-scan-based safety boundary governor for teleoperated robotic haircutting. The proposed method integrates a 3D scan of the mannequin head with a spherical tool-envelope model and regulates boundary-crossing

reference commands before they are executed by the robot. Instead of re-anchoring the operator input, the governor preserves the absolute teleoperation reference and modifies only the inward normal-crossing component of the command by combining boundary projection with normal-component filtering.

The experimental results show that the no-governor stress-test reference can drive the tool envelope into the configured boundary, whereas the proposed governor keeps both the governed command and the measured robot output outside the head-scan boundary in the tested setup. Across five boundary-interaction trials, the prescribed reference input intentionally entered the configured boundary, with a maximum boundary-entry depth of 153.5 mm and 50.3% of samples inside the boundary. In contrast, the governed command and measured output had no inside-boundary samples, with worst boundary distances of 0.1 mm for the commanded trajectory and 0.7 mm for the measured trajectory. These results indicate that the proposed geometric intervention can provide a practical robot-side command-regulation layer for robotic haircutting teleoperation in the tested setup. The present validation is limited to a static mannequin-head setup and a reconstructed head-boundary model. Future work will extend the validation to real heads with motion, hair deformation, scan uncertainty, calibration uncertainty, and tracking uncertainty.

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